Problem Statement:

Design and develop a distributed application to find the coolest/hottest year from the available weather data. Use weather data from the Internet and process it using MapReduce.

In [1]:
```
import pandas as pd
```

In [2]:
```
# Load dataset
df = pd.read_csv("weather.csv")
```

In [5]:
```
# Convert date column
df['Formatted Date'] = pd.to_datetime(df['Formatted Date'], utc=True)
```

In [6]:
```
# Extract year
df['Year'] = df['Formatted Date'].dt.year
```

In [7]:
```
# MAP PHASE
mapped = []
for index, row in df.iterrows():
    mapped.append((row['Year'], row['Temperature (C)']))
```

In [8]:
```
# SHUFFLE & SORT
shuffle = {}
for year, temp in mapped:
    if year not in shuffle:
        shuffle[year] = []
    shuffle[year].append(temp)
```

In [9]:
```
# REDUCE PHASE
result = {}
for year, temps in shuffle.items():
    result[year] = sum(temps) / len(temps)
```

In [10]:
```
# FIND HOTTEST & COOLEST
hottest = max(result, key=result.get)
```

```
coolest = min(result, key=result.get)
```

In [11]:
```
print("Yearly Avg Temp:", result)
print("Hottest Year:", hottest)
print("Coolest Year:", coolest)
```

```
Yearly Avg Temp: {2006: 11.215224505327246, 2005: 0.5777777777777773, 2010: 11.2001764571949, 2007: 12.13467656012176
5, 2008: 12.161818837279903, 2009: 12.269682267884324, 2011: 11.524934043632673, 2012: 11.986823896984417, 2013: 11.9
41016615931, 2014: 12.528227576668824, 2015: 12.31208840690005, 2016: 11.987380925272307}
Hottest Year: 2014
Coolest Year: 2005
```